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Law, Language, and Terminology. Challenges of Generative AI

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Abstract

The paper reflects an interdisciplinary dialogue between legal science, terminology science and computer science aimed at analyzing one specific aspect within the process of digitization of legal operations, in particular via AI: what is the role of labelling and categorization in language and law, and how might legal interpretation be influenced by AI?. We show how a potential challenge in legal AI lies in the obliteration of a certain indeterminacy in legal terms (both concrete and abstract), by arguing that terminology could help controlling the use AI systems make of legal language (via the distinction and systematization of simple terms, concepts, etc.). Terminology-aware generative models would allow more controlled reasoning across languages and jurisdictions and could facilitate explainability by linking each generated statement to a definitional network, thereby exposing the underlying semantic structure of the legal domain. This might help build a trustworthy, explainable AI for law. In this view, terminology could provide an epistemic infrastructure for legal language within an AI operated world, by anchoring generative systems to the conceptual order of law, and by preserving the necessary interplay between determination and indeterminacy that characterises legal language, preventing the risk of current hallucination, over determinacy or the complete loss of penumbra and indeterminacy.

Keywords Law and language · Terminology · Artificial intelligence · Penumbra · Hart · Categorization · Prototype theory

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1 Language, Categorization, and Law

Every thinking and speaking being makes use, on a daily basis, of various processes related both to labelling and categorization: our knowledge of the world, as thinking and speaking beings, is arranged according to categories that ancient philosophy has described in terms of *quantity*, *quality*, etc. (starting, in particular, from Aristotle's *Organon*). The organization of our thought and language according to categories, sub-categories, sets, subsets is necessary in order to associate certain properties to specific groups of objects. For example, once we know that a certain animal falls under the category "birds" we expect it to have feathers and to fly. Problems might emerge when an animal, e.g. a penguin, does have feathers but swims instead of flying: this is a common example used in cognitive linguistics and in prototype theory [46], to prove that when we evaluate the relationship between a particular occurrence and a general category we always have to consider the potential fuzziness of the applicability of general categories to specific cases, the non-homogeneity of such categories, and the necessity to adapt them to changing conditions of reality (so that the incorporation of new instances remains possible without dismantling the precedent categorizations). This is true for ordinary language and for more technical or specific linguistic contexts such as the legal domain.

Other examples of such mechanisms of categorization include the way in which we organize animal species according to the number of legs or to their reproductive system. This allows us to accelerate the processes of memorization and to make inferences according to the category a certain object belongs to (e.g.: birds–fly; fishes–swim, etc.). This may happen consciously, unconsciously or absent-mindedly: we observe the elements making up the situations around us (what to say, what to do, how to judge) and we usually tend to subsume the particular situation within a general level (what people generally say, what people generally do, how people generally judge). Each interpretive community (linguistic, cultural) has its own exemplars (i.e., the specific instantiations that are considered exemplary) and those are considered as prototypical of certain categories (e.g., a sparrow to indicate the category of birds); other instantiations (such as penguins or ostriches) are considered more peripheral in that they do meet the majority of the attributes that the category has and they have a certain proximity with the exemplary instances; nevertheless, they are not considered exemplary (nobody would name a penguin when asked to indicate a bird).

All this leads us to a crucial aspect of the question that we address in the present paper: if this happens with the categorization of our ordinary and scientific knowledge, it certainly happens with our knowledge and use of categorization within the field of law. Like in scientific and in ordinary common-sense reasoning [15, 46], in fact, also in legal reasoning interpreters create connections between particular cases and general rules or principles, so as to organize knowledge, create cause-effect explanations and make inferences [6]. Such a dynamic is widespread in ordinary language, starting from childhood, and it is even more widespread and common within expert contexts like the legal realm, since law works like a "meta-language", i.e., it is structured like a language and in order for it to function it makes use of language, hence it is a sort of "language of language" [4, 5, 40].

Legal practitioners use various operations with the aim of relating specific cases to general rules: among such operations are interpretation and reference to precedent cases. In particular, they follow specific mechanisms and apply certain reasoning methods to judge whether the characteristics of a legal norm apply to particular situations: in most cases, similarity comparison and analogical reasoning play a focal function when establishing whether similar rules apply to different cases [49]. In doing so, legal practitioners (be them lawyers, judges, notaries, etc.) evaluate if the facts of a given case make it meet the criteria for a specific category (i.e., a certain type of contract, a certain crime, a certain fundamental principle of the State like freedom or dignity), therefore in law, as well, categories are not well-delimited sets of instances with clear-cut boundaries. As a matter of fact, many of the terms contained in legal propositions (regulations, conventions, etc.) are potentially contested, much more than terms like 'bird', because all legal systems are rooted in some basic disagreement and hence it is impossible to have the same idea of freedom if there are relevant social distinctions such as class inequalities, or gender asymmetries, or ethnic diversities.

The act of legal interpretation has numerous consequences for the effectiveness of legal texts and for the application of general guidelines to particular situations, both before judgement—when regulations, codes, and conventions are interpreted—and during the adjudicative process itself. What are the concrete objects to which the term 'vehicle' refers? This is a question discussed in the famous US case *Garner v Burr* [1951, 1 KB 31] and then in H.L.A. Hart's *Positivism and the Separation of Law and Morals*. Through the example of a rule excluding vehicles from the park, Hart differentiated the straightforward applications of a rule (what he called the rule's "core") from the harder cases at a rule's edge (what he called "penumbra", i.e., the area of indeterminacy of meaning). In the presentation of open texture of law, i.e. a "texture" constantly under interpretation, Hart uses the term 'penumbra' to refer to what in prototype theory is called 'periphery' (the area where things could be defined in more than one way), a term mentioned above regarding the example of bird categorization in zoology.

In law, the relationship between a clear semantic center—whether it concerns terms referring to concrete objects, such as 'vehicle', or abstract concepts, such as 'freedom'—and a peripheral or penumbral area raises several issues related to analogy, similarity assessment, and inferential reasoning (all issues that we cannot address in this paper). With terms referring to abstract objects/concepts such as free speech, dignity, and discrimination, the level of basic disagreement can be higher, because their meaning depends on entire visions of the world. If we consider, for instance, the First Amendment to the US Bill of Rights,[1] the notion of free speech has a rather broad realm of application: within such a realm, judges decide cases involving false speech (*Debs v. US*, 1919), "fighting words" (*Cantwell v. Connecticut*, 1940), freedom of assembly and public forums (*Schneider v. New Jersey*, 1939), symbolic speech (*US v.*

[1] "Congress shall make no law respecting an establishment of religion or prohibiting the free exercise thereof; or abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the Government for a redress of grievances".

O'Brien, 1968), compelled speech (*Wooley v. Maynard*, 1977), school speech (*Morse v. Frederick*, 2007), obscenity (*US v. Stevens*, 2010) etcetera [13].

The act of naming, labelling, categorizing, and the related act of interpreting certain events, objects, facts, phenomena within the legal realm is never straightforward or purely neutral: if legal reasoning is built on syllogism, we can say that no premise (be it major or minor) is ever perfectly ready. Legal interpretation often requires, beyond deduction and induction, also abductive reasoning [6].

Against this background, we believe that further discussions on this topic could be developed today, starting in particular from some core assumptions of terminology and the digitalization of various legal operations via AI software (e.g., contract drafting, predictive justice, legal data retrieval). The increasing use of such software by legal practitioners (both private and public) is creating a series of issues related to how naming, labelling, and categorization have traditionally functioned in legal practice and how they might operate in the future (i.e., interpretation is becoming more hybrid—both human and computational). Several issues arise in this context, including: what happens with the conversational implicatures and the implicit meanings [8, 43], which are fundamental in law, and what happens with the "intentional" nature of legal concepts, such as those above mentioned but also, for instance, privacy or property? [1] Legal language is indeed often complemented by implicatures, i.e., by themes or assertions that are implicit within utterances, either more specific or more general and that relate to legal systems. These are, for example, general principles or *regulae iuris*: as stated by the Italian Supreme Court in decision 2003/8827, there are certain basic principles that are implicit within legal systems and are nevertheless necessary, mandatory, and should be fully respected despite their implicit nature, as derived from a joint interpretation of several provisions of the Italian Constitution. Conversational implicatures are implied meanings that are underlying in communication: the speaker or interpreter suggests something beyond the literal meaning, relying on the background, the context or on shared knowledge. Like other emergent properties (e.g., understanding or consciousness), implicatures are difficult to be incorporated in computing machines, because they are contextual and depend on the interpretation of each case and because there are some basic differences between human mind and computer that are impossible to overcome.

The "penumbra" mentioned by Hart is indeed a vital section of the meaning of legal terms; because of indeterminacy, penumbra allows adjustment of meanings to singular cases. The question we address in this paper is how such indeterminacy, which is vital for the interchanges between legal propositions and extra-legal values, can be preserved through the digitalization of legal operations, with specific regard to the risks and potentialities of new AI software. A certain amount of indeterminacy is necessary. How can it be preserved with actual technologies? In other words, how can we prevent the erosion of the penumbra or its "hyper-illumination" (excessive clarification of the indeterminacy)? To address this question, we argue that the systematization of terms as developed within terminology science—namely, the identification and structuring of domain-specific concepts, their conceptual relations, and the terms used to designate them—may be helpful. It is this specific contribution of terminology that we are going to consider in the central section of the paper [14].

All this is further complicated by the fact that, at times, guidance on how a term should be applied comes not from a norm but from another case that effectively functions as one. Such a phenomenon can be defined as “exemplarity” and it is extremely important in law, as the case above mentioned (*Garner v Burr*) or for instance *Obergefell v Hodges* (2015) or *Roe v Wade* (1973) can prove [12]. Once again, this is a peculiarity of the legal realm that may be undermined by the growing reliance on AI software to perform legal tasks. All phenomena depending on categorization might undergo changes because of the logic of AI.

In essence, to conclude on the first sequence of remarks, it might be interesting to connect the question of how to maintain the aura of indeterminacy, vagueness, etc., in legal terms (which is vital, because profound disagreement is the basis of correctly “practiced” law, since full agreement or consensus on sensitive, indeterminate, etc. issues is impossible) as well as certain dynamics related to the “exemplarity” of specific cases—with the methodological approach proposed by terminology. The fundamental question we aim to flesh out with this paper is therefore to what extent terminology can aid the algorithmic transition in law, and on such transition, we focus in the third section of the paper.

2 Terminology Science and Terminology Work

Section 1 has shown that labelling and categorization in law operate through meanings that have both a definitional nucleus and a necessary penumbra. This openness is not a defect but a functional resource of legal reasoning, which must connect rules to cases across shifting factual and evaluative contexts. The methodological problem, especially in view of the increasing hybridization of legal practice with digital and generative tools, is how to preserve such openness while making meaning sufficiently explicit to support reasoning, adjudication, and—where appropriate—computational assistance. Terminology science and terminology work can help in addressing this problem by modelling specialized knowledge at the level at which law operates and by documenting their designations, relations, sources, and usages in a way that renders stability and contestation jointly visible.

2.1 The Discipline: Scope and Core Elements

Terminology science deals with the organization, representation, and transfer of specialized knowledge. Although the discipline began to take shape around the 1930s, its epistemological status remains a topic of considerable debate. Looking back at its origins, it is notable that the “spiritual fathers of modern terminology” were three engineers—Eugen Wüster (Austria), Ernst Drezen (Latvia), and Dmitrij Lotte (Russia)—who advocated for the necessity of clear and effective specialized communication in both professional and dissemination contexts [42]. Among them, Wüster stood out for developing the General Theory of Terminology on a philosophical-linguistic and logical basis, with the primary intent of organizing specialized knowledge and facilitating its correct transmission [57]. Since the 1990s, this perspective has faced opposition within parts of the linguistic community [7], with strands that either sought to reclaim terminology as a branch of linguistics or relegated it to specialized lexicology [2]. Against these reductions, a substantial body of work argues that terminology is an autonomous science precisely because it examines and interrelates both the conceptual and the linguistic dimensions of specialized knowledge [16, 36, 51]. In practice, terminology work considers at once the conceptualizations of domain experts with respect to specific objects (e.g., oncologists' knowledge about breast cancer) and the discourses they produce to transmit that knowledge [8]. Because of this peculiar attitude, by bridging methods and purposes, terminology might contribute—in our view—to facilitate labelling and categorization within law, a field where terms span both the level of concepts and the level of concrete objects, often within the same sentence.

A crucial component of the field of terminology concerns the design and development of resources that make a domain's terminology accessible in print and, increasingly, digital media [49]. This is particularly relevant from the point of view of law, where daily practice is increasingly conducted through software, quite often AI software for data retrieval and document drafting, as well as for judgment prediction.

At the international level, ISO TC 37/SC 3 gathers experts from academia and industry to standardize models and processes for the effective management of terminology resources. These standards focus not simply on words, but on how units of knowledge (i.e. "concepts") should be represented and exchanged in digital environments. In this framework, interoperability is promoted by adopting a concept-oriented structural meta-model, for example ISO 16642: 2017 (TMF) [27], which organises information around conceptual entities and their relations. A concept-oriented approach starts from the legal notion itself (e.g. the concept of "contract" as a binding agreement) and then associates it with the relevant linguistic expressions in each language (e.g. contract in English, contratto in Italian, contrat in French). The focus is on the legal meaning and its attributes (parties, obligations, validity conditions, etc.) rather than on the word form alone. This differs from the lemma-oriented structuring typical of lexicographic resources [29], where the starting point is the word (the lemma) and usage descriptions linked to it (e.g. dictionary entries for "contract," including figurative uses like "to contract a disease"). While lexicography organises information around words, terminology standards organise it around concepts, a distinction that becomes crucial when terms need to be exchanged, modelled or processed by computational systems.

The envisaged user groups depend on domain needs. In medical terminology, for instance, the construction of resources is closely aligned with science popularization and health literacy, where accessibility and cognitive facilitation for non-experts are paramount [54, 56]. As far as legal terminology is concerned, as above mentioned, terms can be, within the same proposition (e.g. an article of the Civil code), pretty specific (e.g., 'emphyteusis' is an example of a term used merely to indicate a contract in land law) and pretty popularized and related to common sense perception of certain values (e.g. 'justice', 'freedom', 'dignity'). The legal domain is therefore particularly interesting from this perspective because it is, as we said already, a "language of a language", i.e., a specialistic language that incorporates ordinary language terms by attributing to them a certain value.

Another key constituency comprises language professionals—specialized translators, interpreters, and technical communicators—who rely on interoperable, reusable multilingual resources to ensure accuracy and consistency in high-stakes contexts [5].

Clarifying the discipline's core entities helps to explain its autonomy. While several semiotic models have been proposed [37], an adaptation of the Ogden and Richards [33], triangle remains widespread, distinguishing object, concept, and designation. The theoretical treatment of objects and concepts draws on philosophy of language: objects are the raw material for concept formation, and conceptual analysis examines how experts structure those objects into intelligible units [42, 52]. In ISO 704: 2022 [30] terms, an object is anything perceivable or conceivable and is characterized by properties; through conceptualization, experts abstract these properties and transform them into characteristics, whose unique combination constitutes a concept. The literature debates the nature of the concept and here are three attempts to define such a nature: Wüster's "unit of thought" [57], Dahlberg's "unit of knowledge" [18], Temmerman's "unit of understanding" [50]. What most terminologists agree on is that the concept is extra-linguistic, intersubjective, and revisable, and that only concept-level modelling supports durable, sharable knowledge organization. The designation is the representation of a concept by a sign within a subject field [27]. At the same time, debates on the linguistic nature of the term continue—variously framed as "linguistic sign" (Saussurean), "denomination" (onomastic tradition), or "lexical form" (lexical semantics) [19, 35, 55], but the concept-orientation of the resource remains the decisive organizational principle [53].

2.2 Representing Necessary Indeterminacy in Terminology Resources

The conclusion of the first paragraph on the interconnection between law and language clarified that it would be useful to integrate indeterminacy within terminology resources for the legal domain. A concept-oriented entry makes this possible by co-locating the elements that stabilize concepts with those that keep them open to principled disagreement. In line with ISO 1087 [27] and ISO 704 [30], and with the practice of legal terminology and legal translation [3, 9, 38, 47], the entry treats the concept as an intersubjective, revisable unit of knowledge, while designations function as representational forms of knowledge. Picht's synthesis is instructive here: only stabilized knowledge—definitions, relations, scope, provenance—belongs in thermographic storage, but stabilization does not entail closure; rather, it is the condition for documenting contestation with sources and dates [41, 42].

Each entry begins with a persistent concept identifier that anchors information specific to jurisdictions and time periods. It then presents, side by side, two definitional layers. The authoritative definition reproduces (with citation) the wording from legislation, case law, or other binding sources. The working (intentional) definition abstracts necessary characteristics within the current concept system. Immediately thereafter, scope and applicability specify jurisdiction, area of law, and temporal validity (in force / historical), together with references to general principles implicated. Representational forms link the concept to designations: preferred and admitted terms, multiword expressions, and, where appropriate, doctrinal tests or standards that shape application (e.g., proportionality). Relational placement situates the concept within a mixed system, combining generic–specific hierarchies with ontological relations (part–whole, sequential, causal, interdependence). Usage notes articulate prototypical instances, recognized borderline cases, and competing doctrinal readings; uncertainty/provenance metadata qualify each assertion with authority level, source, and date, allowing divergence to be recorded rather than flattened. Finally, in multilingual or comparative settings, cross-lingual mappings are stated as equivalence types (full, partial, near-equivalence, non-equivalence) with justifications, reflecting the concept-led alignment long emphasized in legal linguistics [44].

3 Retrieval-Augmented Generation in Legal AI: Risks and Evaluation

The increasing integration of artificial intelligence into legal practice invites a re-examination of the relationship between law, language, and categorization in computational terms [58]. If terminology provides the conceptual scaffolding through which legal meaning can be structured and communicated, AI systems now operate as new mediators of that structure and for this reason we believe interdisciplinary work between legal science, terminology science, and computer science can be useful to identify potential misuses of technology and to construct a system of checks and balances necessary to guarantee a proper use of terms by computational instruments within fields of expert knowledge. Among various technological paradigms, Retrieval-Augmented Generation (RAG) represents a turning point: by coupling large language models with explicit access to legal sources, it promises to reconcile the indeterminate, vague and sometimes ambiguous nature of legal language with the need for machine-assisted precision [23], an exactness that, in law, is never purely technical but emerges from interpretive practices and socio-legal contexts. Yet, this hybridisation of retrieval and generation also exposes critical tensions, between statistical inference and conceptual determination, between automation and interpretive openness. Nevertheless, it is precisely this productive tension that makes RAG potentially transformative for law: the model's generative capacity can be constrained, steered or even enriched by normative sources, while those same sources must be interpreted within evolving socio-legal contexts rather than merely "retrieved." The following section analyses these tensions, outlining some of the main epistemic risks and evaluative challenges raised by RAG in the legal-terminological domain.

3.1 From Data-Driven to Knowledge-Augmented Legal AI

Large language models (LLMs) have recently transformed much knowledge-intensive domains, and law is no exception [33]. From contract drafting to legal question answering, such systems now assist professionals in producing, summarizing, or interpreting texts that previously required highly specialised expertise. However, the performance of these models remains essentially statistical: they capture lexical and syntactic co-occurrences but do not possess an internalised representation of legal categories, definitional structures, or normative values [34]. Their ability to reproduce legal discourse is therefore not equivalent to legal understanding and hence the main problem we address here, also considering the remarks made on the specificity of legal terminology, is an "erosion" of the penumbra Hart described, caused by an excessive illumination (i.e., definition/determination of the meaning) that would be conter-productive in regard to the necessary indeterminacy characterizing legal terms, an indeterminacy that is directly linked to the preservation of basic disagreement.

Retrieval-Augmented Generation (RAG) has emerged as an attempt to bridge this epistemic gap. Instead of generating responses solely from parameters pre-trained on massive and opaque corpora, a RAG system couples a retrieval module, which fetches contextually relevant texts such as statutes, case law, or doctrinal sources, with a generation module that conditions its output on those retrieved materials. In other words, RAG operationalises a hybrid reasoning mode that joins statistical inference with explicit reference to authoritative texts. In this sense, it performs a kind of computational hermeneutics: meaning is not generated *ex nihilo* but through contextual anchoring to legal language itself [31, 45].

Such architectures are particularly appealing in law, where traceability and justification are essential. They promise to combine the flexibility of generative models with the epistemic discipline of source-based reasoning. Yet the quality of the retrieval stage, the corpus searched, the ranking algorithms, and the way snippets are integrated into the prompt, directly affect the validity of the generated argument. The legal soundness of a RAG system therefore depends less on sheer model size than on the design of its knowledge interface.

3.2 Legal Terminology and the Challenge of Referential Precision

As discussed earlier, legal reasoning is grounded in acts of naming, labelling, and categorising: law is basically a human practice functioning through language. Terms in law are rarely univocal: their meaning emerges from systemic relations within a jurisdiction and from their interpretative history [44]. Even apparently concrete terms, such as vehicle, weapon, employee, are subject to contextual negotiation. Abstract notions such as dignity, discrimination, or privacy are even more open to reinterpretation.

In multilingual or cross-jurisdictional contexts like the European Union, these subtle differences multiply and with them also the potential risks, as reflected in EU-level efforts both to digitalise justice (https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52020DC0710; https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/digitalisation-justice_en) and to regulate AI, including in justice-related settings (for example, AI Act https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng). A term like 'property' has distinct semantic fields in common-law and civil-law traditions; responsibility or liability has different normative connotations in English, French, German or Italian (each of these concepts refers as well to different legal traditions and cultures). LLMs trained on mixed data tend to blur such boundaries, producing paraphrases or analogies that are linguistically plausible but that at the same time might be misleading from the point of view of legal interpretation and application. Moreover, as above mentioned, current AI systems tend to find, via statistics, the most recurrent terms associated with other terms (e.g., GenAI). Such mechanism might not only erode or "hyper-illuminate" the penumbra of legal terms, which should instead remain partially indeterminate, but could also create new associations between terms (because they are frequent in certain fields of law) and make them become "regular", thus confusing the connections between signs and meanings even more. This is in particular the type of problems we are focussing on and on which we shall test our theses (the results of future collaborations shall be published in a different series of papers).

The empirical work we propose specifically targets this phenomenon. Our aim is to construct controlled multilingual datasets reflecting legally valid concept networks (e.g. different types of responsibility or property). Ideally, we would build small, purpose-built corpora centred on a limited set of legal concepts (for instance different types of responsibility or property), where each occurrence is drawn from authoritative legal sources (legislation, case law, doctrinal texts) and manually annotated with the corresponding concept, jurisdiction, and language. These datasets will not simply be parallel texts, but concept-oriented collections: for each target notion we will identify its subtypes, conditions of application, and typical collocations in each legal system, and represent them as a network of concept nodes linked by defined relations (broader/narrower term, cause–effect, role, etc.). This network constitutes what we call an expert-validated conceptual structure: a reference graph co-constructed by legal scholars and terminologists, which encodes how concepts are distinguished and related in a given legal tradition and serves as a gold standard against which model behaviour can be evaluated. On this basis, we plan to test how a RAG system reorganises, modifies, or distorts these relations when generating answers. Concretely, we will query the system with prompts designed to elicit definitions, paraphrases, and analogies for the target concepts (e.g. "Explain strict liability in Italian administrative law" or "Compare propriété and ownership in EU consumer law"), while constraining retrieval to the controlled datasets. The model's outputs will then be mapped back onto the expert-validated conceptual structures: for example, we will check whether subtypes are correctly distinguished, whether conditions of application are preserved, and whether analogies respect jurisdictional boundaries. Such an experiment would provide computer scientists with benchmarks to evaluate semantic distortions and offer terminologists a systematic diagnostic tool to identify where and how legal concepts are being "misaligned" by algorithmic inference.

RAG provides a partial safeguard by grounding generation in retrieved documents from specific jurisdictions. When coupled with structured terminological resources, glossaries, ontologies, or linked data repositories, it can enhance referential precision [20] and it might thus reduce the risks of confusion mentioned above, which are the other side of the "efficiency" coin in using such systems for legal operations. However, this depends on the quality, granularity, and interoperability of those terminological assets, that is, on how accurate the concepts are, how finely their legal attributes are specified, and how easily this information can be exchanged between different systems or databases. In particular, granularity refers to the level of detail with which a concept is represented: for instance, whether the term "liability" is merely labelled in general terms, or whether its specific subtypes (strict liability, fault-based liability, vicarious liability, administrative liability, etc.) are distinguished together with their conditions, scope, and jurisdictional constraints. If retrieval sources are unbalanced, outdated, or lacking explicit metadata, even a sophisticated model may conflate meanings across systems.

Hence, the integration of terminology management into the technical pipeline becomes very useful. Terminological databases following the FAIR principles (Findable, Accessible, Interoperable, Reusable) can function as curated retrieval spaces where definitions, conceptual relations, and references are explicitly modelled [51]. When a RAG system draws from such repositories, it not only grounds generation in reliable data but also inherits the underlying conceptual architecture of the legal domain. In this perspective, terminology ceases to be a peripheral linguistic concern and becomes the structural backbone of legal AI.

3.3 Evaluating Legal RAG Systems: Transparency and Controllability

In the RAG pipeline, between retrieval (locating the relevant legal sources), ranking (prioritising what resource appears more pertinent or reliable), and fusion (combining those sources with the model's generated text), each computational layer introduces its own biases. Evaluating such systems thus requires moving beyond quantitative accuracy to a broader notion of interpretive adequacy.

Relevant dimensions of evaluation include:

- Faithfulness, i.e. the correspondence between generated content and retrieved sources;
- Terminological consistency, the correct and jurisdictionally appropriate use of legal terms in line with reference standards such as EuroVoc (https://tinyurl.com/mw3adc2k) or IATE (https://iate.europa.eu/)
- Explainability, understood as the ability to trace outputs to specific retrieved passages;
- Cross-lingual coherence, the stability of conceptual mappings across EU legal languages;
- Normative alignment, the extent to which the system respects the hermeneutic openness inherent in legal discourse.

Current benchmarks such as LexGLUE [11] or EUR-Lex (https://eur-lex.europa.eu/) provide useful starting points, but they remain largely quantitative whereas, as we have shown, the question at stake in law remains mainly qualitative. Human evaluation by legal and linguistic experts remains indispensable to assess how the system handles ambiguity, implicature, or competing interpretations—dimensions that define legal reasoning itself.

3.4 Risks: Hallucination, Over-Determinacy, and the Loss of Indeterminacy

Even with retrieval, generative models can still "hallucinate" by fabricating citations, merging distinct legal principles, or re-contextualising rules in unintended ways [17]. In legal settings, such errors are not merely factual but normative: they may distort the balance between general principles and individual cases. A subtler and more structural risk is over-determinacy. By producing apparently definitive answers, automated systems can erase the interpretive indeterminacy that is both inevitable and desirable in law [14, 26]. The danger is not only technical but epistemological: when meaning is stabilised by algorithmic authority, the pluralism of legal interpretation may be weakened.

Therefore, evaluation criteria should include epistemic humility. A responsible legal RAG system should signal uncertainty, delimit the scope of its claims, and provide explicit references enabling human verification. Rather than replacing legal interpretation, such systems should make visible the zones of interpretive flexibility that define legal discourse and evidentiate the terms on which there might be disagreement, hence distinguishing the way terms are treated within contexts.

3.5 Frontiers: Towards Terminology-Aware Generative Models

With this paper we have tried to show how a potential frontier in legal AI lies in the deeper integration of terminological theory into model design. Current RAG systems use terminology mainly as an external resource for retrieval. Future architectures could embed terminological relations directly within the model's representational space [20, 32]. Graph-based embeddings [23], for instance, can encode hierarchical and associative relations (generic/specific, part/whole, oppositional) that mirror the conceptual systems developed by terminologists [38].

Such terminology-aware generative models would allow more controlled reasoning across languages and jurisdictions. They could also facilitate explainability by linking each generated statement to a definitional network, thereby exposing the underlying semantic structure of the legal domain. This hybridisation of symbolic and statistical methods represents a concrete step towards trustworthy, explainable AI for law [51].

In this view, terminology constitutes the epistemic infrastructure of legal AI. It anchors generative systems to the conceptual order of law, preserving the necessary interplay between determination and indeterminacy that characterises legal language. The collaboration between lawyers, terminologists, and engineers is thus not ancillary but constitutive. Law provides the normative and interpretive framework; terminology supplies the conceptual scaffolding; AI contributes the computational means to operationalise both. Only through this triadic cooperation can generative technologies serve, rather than subvert, the hermeneutic rationality of legal systems. Future empirical studies will be devoted to testing the theses discussed in the paper.

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References

1. Arapinis, Alexandra, and Angela Condello. 2016. The intensionality behind legal concepts and their extensional boundaries: Between conventionalism and interpretivism. *Ratio Juris* 29:439–459.
2. Bergenholz, Henning, and Sven Tarp. 1995. *Manual of specialised lexicography*. Amsterdam/Philadelphia: John Benjamins Publishing Company.

3. Biel, Łukasz. 2019. Theoretical and methodological challenges in researching EU legal translation. Legal translation. *Current issues and challenges in research, methods and applications*: 25–39.
4. Bobbio, Norberto. 1950. Scienza del diritto e analisi del linguaggio. *Riv. trim. di diritto e procedura civile* 2.
5. Bowker, Lynne. 2015. *Terminology and translation*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
6. Brewer, Scott. 1996. Exemplary reasoning. Semantics, pragmatics, and the rational force of legal argument by analogy. *109 Harvard Law Review* 5:923–1028.
7. Campo, Ángel. 2012. *The reception of Eugen Wüster's work and the development of terminology*. Université de Montreal (Canada), PhD thesis.
8. Canale, Damiano. 2019. *En Busca de lo Implícito. Ensayos Sobre Razonamiento e Interpretación en el Derecho*. Bogotà: Universidad Externado.
9. Cao, Debora. 2007. *Translating law* (Vol. 33). Multilingual Matters.
10. Carvalho, Sérgio, Costa, Rute and Roche, Christophe. 2016. LESS Can Indeed Be More: Linguistic and Conceptual Challenges in the Age of Interoperability. In: *12th International conference on Terminology and Knowledge Engineering (TKE 2016). Term Bases and Linguistic Linked Open Data*. Copenhagen Business School.
11. Chalkidis, Ilias, Jana, Abhik, Hartung, Daniel, Bommarito II, Michael J., Androutsopoulos, Ion, Katz, Daniel Martin and Aletras, Nikolaos. 2022. LexGLUE: A Benchmark Dataset for Legal Language Understanding in English. In: S. Muresan, P. Nakov, & A. Villavicencio (Eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 4310–4330). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.297.
12. Condello, Angela. 2018. *Between Ordinary and Extraordinary. The Normativity of the Singular Case in Art and Law*. Leiden: Brill.
13. Condello, Angela and Arapinis, Alexandra. 2017. Between similarity and analogy. Rethinking the role of prototypes in law and cognitive linguistics. In: *The pragmatic turn in law. Inference and Interpretation*, eds. J. Giltrow e D. Stein, 167–185. Berlin: De Gruyter Mouton. https://doi.org/10.1515/9781501504723-007.
14. Condello, Angela and Di Nunzio, Giorgio Maria. 2024. Legal Science and Computer Science: A Preliminary Discussion on How to Represent the "Penumbra" Cone with AI. In: *Proceedings of the Workshop on DeTermIt! Evaluating Text Difficulty in a Multilingual Context @ LREC-COLING*, 179–184.
15. Cornell Way, Eileen. 1991. *Knowledge representation and metaphor*. Heidelberg: Springer.
16. Costa, Rute. 2013. Terminology and specialised lexicography: Two complementary domains. *Lexicographica* 29:29–42. https://doi.org/10.1515/lexi-2013-0004.
17. Dahl, Molly, Varun Magesh, Mirac Suzgun, and Daniel E. Ho. 2024. Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis* 16 (1): 64–93.
18. Dahlberg, Ingetraut. 1976. Über Gegenstände, Begriffe, Definitionen und Benennungen: Zur möglichen Neufassung von DIN 2330. In: *Muttersprache* 2/1976.
19. Depecker, Loïc. 2002. *Entre signe et concept: éléments de terminologie générale*. Paris: Presses Sorbonne Nouvelle.
20. Di Nunzio, Giorgio Maria. 2025. Terminology-augmented generation (TAG): Foundations, use cases, and evaluation paths. *Journal of Digital Terminology and Lexicography* 1 (1): 97–104. https://doi.org/10.25430/pupj.jdtl.1752566034.
21. Di Nunzio, Giorgio Maria and Condello, Angela. 2024. An Interdisciplinary Approach to Legal Terminology: Challenges of the "Algorithmic Turn" in Legal Science. In *3rd Workshop on Augmented Intelligence for Technology-Assisted Review Systems (ALTARS 2024): Evaluation Metrics and Protocols for eDiscovery and Systematic Review Systems*.
22. Endicott, Timothy. 2022. Law and Language. *The Stanford Encyclopedia of Philosophy*, Edward N. Zalta (ed.). https://plato.stanford.edu/archives/spr2022/entries/law-language/.
23. Gao, Yujia, Yifan Xiong, Xiaowei Gao, Kai Jia, Jiaqi Pan, Yun Bi, Yuntao Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-augmented generation for Large Language Models: A survey. *arXiv*. https://doi.org/10.48550/arXiv.2312.10997.
24. Hart, Herbert Lionel Adolphus. 1958. Positivism and the Separation of Law and Morals. In 71 *Harvard Law Review*. 593: 606–15.
25. Hart, Herbert Lionel Adolphus. 1961. *The Concept of Law*. Oxford: Clarendon.

26. Hernández, Guillermo. 2022. Law's determinability: Indeterminacy, interpretative authority, and the international legal system. *Netherlands International Law Review* 69 (2): 191–219.
27. ISO 1087. 2019. *Terminology Work and Terminology Science – Vocabulary – Part 1: Theory and application*. Geneva: International Organization for Standardization.
28. ISO 16642. 2017. *Computer Applications in Terminology – Terminological markup framework*. Geneva: International Organization for Standardization.
29. ISO 24613-1. 2024. *Language resource management – Lexical Markup Framework (LMF) – Part 1: Core model*. Geneva: International Organization for Standardization.
30. ISO 704. 2022. *Terminology Work — Principles and Methods*. Geneva: International Organization for Standardization.
31. Jamesju. 2024, December 4. Intro to retrieval-augmented generation (RAG) in legal tech. *Thomson Reuters Law Blog*. https://legal.thomsonreuters.com/blog/retrieval-augmented-generation-in-legal-tech/.
32. Lackner, Anna, Ana Vega-Wilson, and Christoph Lang. 2025. An evaluation of terminology-augmented generation (TAG) and various terminology formats for the translation use case. *Journal of Digital Terminology and Lexicography* 1 (2): 31–47. https://doi.org/10.1234/jdtl.v1i2.3.
33. Lai, Jie, Wei Gan, Jie Wu, Zhengqui Qi, and Philip S.. Yu. 2024. Large language models in law: A survey. *AI Open* 5:181–196. https://doi.org/10.1016/j.aiopen.2024.09.002.
34. Loru, Edoardo, Jacopo Nudo, Nicola Di Marco, Alessio Santirocchi, Riccardo Atzeni, Michela Cinelli, Valentina Cestari, Camille Rossi-Arnaud, and Walter Quattrociocchi. 2025. The simulation of judgment in LLMs. *Proceedings of the National Academy of Sciences* 122 (42): e2518443122. https://doi.org/10.1073/pnas.2518443122.
35. L'Homme, Michel C. 2020. *Lexical semantics for terminology: An introduction*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
36. Meyer, Ingrid. 1993. Concept Management for Terminology: A Knowledge Engineering Approach. Standardizing Terminology for Better Communication: Practice, Applied Theory, and Results. Ed. Strehlow, R, and Wright, S. 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428–2959: ASTM International.
37. Myking, J. 2001. Sign models in terminology: tendencies and functions. *LSP and professional communication (2001–2008)* 1(2).
38. Nuopponen, Anna. 2022. Chapter 3. Conceptual relations: From the General Theory of Terminology to knowledge bases. In: *Theoretical Perspectives on Terminology: Explaining terms, concepts and specialized knowledge*. John Benjamins Publishing Company: 63–86.
39. Ogden, Charles Kay, and Ivor Armstrong Richards. 1927. *The Meaning of Meaning: A Study of the Influence of Language upon Thought and of the Science of Symbolism*. Brace: Harcourt.
40. Oppenheim, Felix. 1944. Outline of a logical analysis of law. *Philosophy of Science* 11 (3): 142–160.
41. Picht, Helmut. 2009. The seven pillars of terminology. *Terminologija* 16:8–22.
42. Picht, Helmut. 2011. The science of terminology: History and evolution. *Terminologija* 18:6–26.
43. Poggi, Francesca. 2020. *Il modello conversazionale. Sulla differenza tra comprensione ordinaria e interpretazione giuridica*. Pisa: ETS.
44. Prieto Ramos, F. 2021. Translating legal terminology and phraseology: Between inter-systemic incongruity and multilingual harmonization. *Perspectives* 29 (2): 175–183. https://doi.org/10.1080/0907676X.2021.1849940.
45. Raditya, Kasuma and Sayed, Pantho. 2025, April 3. *Retrieval-augmented generation (RAG): Towards a promising LLM architecture for legal work?* Harvard Journal of Law & Technology. https://jolt.law.harvard.edu/digest/retrieval-augmented-generation-rag-towards-a-promising-llm-architecture-for-legal-work.
46. Rosch, Eleanor. 1973. Natural categories. *Cognitive Linguistics* 7:532–547.
47. Šarčević, Snježana. 1997. New approach to legal translation.
48. Scarpelli, Ugo. 1948. Scienza del diritto e analisi del linguaggio. *Rivista del diritto commerciale* 5–6: 212 s.
49. Schmitz, Klaus-Dirk. 2015. Terminology and localization. *Handbook of Terminology*, 1, Amsterdam/Philadelphia: John Benjamins Publishing Company: 452-464.
50. Sunstein, Cass R. 1993. On analogical reasoning. *Harvard Law Review* 106 (3): 741–791.
51. Temmerman, Rita. 2000. *Towards New Ways of Terminology Description: The Sociocognitive Approach*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
52. Vezzani, Federica. 2022. *Terminologie numérique: Conception, représentation et gestion* : 238. *Peter Lang International Academic Publishers*. https://doi.org/10.3726/b19407.

53. Vezzani, Federica. 2023. La connotation du vocabulaire somatique: Une étude de cas comparative bilingue en oncologie. *Meta* 68 (1): 97–118. https://doi.org/10.7202/1106329ar.
54. Vezzani, Federica, and Rute Costa. 2024. Variation in psychopathological terminology: A case study on Body Dysmorphic Disorder. *Terminology. International Journal of Theoretical and Applied Issues in Specialized Communication* 30 (1): 81–106. https://doi.org/10.1075/term.00078.vez.
55. Vezzani, Federica, and Giorgio Maria Di Nunzio. 2020. Methodology for the standardization of terminological resources: Design of TriMED database to support multi-register medical communication. *Terminology. International Journal of Theoretical and Applied Issues in Specialized Communication* 26 (2): 265–297. https://doi.org/10.1075/term.00053.vez.
56. Warburton, Katherine. 2021. *The Corporate Terminologist*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
57. Wermuth, Christiane and Verplaetse, Herman. 2019. Medical terminology in the Western world. *Handbook of terminology*, *1*. Amsterdam/Philadelphia: John Benjamins Publishing Company. https://doi.org/10.1075/hot.2.05wer.
58. Wüster, Eugen. 1979. *Einführung in die Allgemeine Terminologielehre und Terminologische Lexikographie*. Wien, New York: Springer Verlag.
59. Zafar, Ammar. 2024. Balancing the scale: Navigating ethical and practical challenges of artificial intelligence (AI) integration in legal practices. *Discover Artificial Intelligence* 4 (1): 27. https://doi.org/10.1007/s44163-024-00121-8.